

ISO26262 Functional Safety Report

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Development Method: **BDD**

Field	Value
Document Template	safety_report
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Requirement	vehicle
Related ISO Parts	Part 11 - Semiconductors Part 6 - Software Development Unknown
Referenced Pages	125, 128, 140, 145, 36, 39, 71, 9

ISO26262 Evidence

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 9

Score: 0.4119

Summary: ■ Road vehicles — Functional safety — Part 11: Guidelines on application of ISO 26262 to semiconductors 1 Scope This document is intended to be applied to safety-related systems that include one or more electrical and/or electronic (E/E) systems and that are installed in series production road vehicles, excluding mopeds. This document does not address unique E/E systems in special vehicles such as E/E systems designed for drivers with disabilities. NOTE Other dedicated application-specific safety standards exist and can complement the ISO 26262 series of standards or vice versa. Systems and their components released for production, or systems and their components already under development prior to the publication date of this document, are exempted from the scope of this edition. This docu

Part: Part 11 - Semiconductors

Clause: 5.5.2

Page: 128

Score: 0.3445

Summary: ■ ISO 26262-11:2018(E) Figure 28 — General relationship between sensor and transducer Like other E/E elements, a sensor can be made up of parts and subparts, and be of varying complexity. EXAMPLE 2 A semiconductor component with analogue output consisting of a transducer and amplifier. EXAMPLE 3 An element consisting of housing, a sensor IC with digital signal processing and digital output, required external components (e.g. resistors, capacitors) and a connector which interfaces to a wiring harness (see Figure 29). In this example, both the sensor IC and other elements can be classified as sensors but exist at different levels of hierarchy. NOTE 2 The term 'transducer' in this sub-clause refers specifically to those transducers that are fabricated using semiconductor process technology, i

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Clause: Unknown

Page: 39

Score: 0.3363

Summary: ■ ISO 26262-11:2018(E) NOTE 1 In cases where the number of equivalent transistors N is not listed in the failure rates families tables 1, 2 or 3 of SN 29500-2:2010 and when possible the user can use an interpolation or extrapolation method to determine the equivalent λ_{ref} and $\theta_{vj,1}$ (virtual junction temperature) values. EXAMPLE For "microprocessors and peripherals, microcontrollers and signal processors" family as defined in SN 29500-2:2010, Table 2, the following interpolation example is done to determine λ_{ref} and $\theta_{vj,1}$ values. Assuming a microcontroller with 500 K gates the calculation of the λ_{ref} could be done using the following steps: ■ 1st step: translating λ_{ref} values from Table 2 to a same virtual reference temperature $q_{vj,1}$ for example 90 °C by using the temperature dependent fact

Part: Part 6 - Software Development

Clause: 5.4.2

Page: 125

Score: 0.3356

Summary: ■ ISO 26262-11:2018(E) ■ heterogeneous multi-cores components which have non-identical PEs, typically with different Instruction Set Architecture (ISA). EXAMPLE Figure 27 shows a diagram of a generic homogeneous dual-core system, with CPU-local level 1 caches, and a shared, on-die level 2 cache. Figure 27 — Generic diagram of a dual-core system 5.4.2 Implications of ISO 26262 series of standards for multi-core components 5.4.2.1 Introduction This sub-clause provides guidance for cases where safety requirements — previously allocated to multiple components — are now allocated to a multi-core. 5.4.2.2 Clarifications on Freedom from interference (FFI) in multi-core components If in a multi-core context multiple software elements with different ASIL ratings coexist, a freedom from interference

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Clause: Unknown

Page: 36

Score: 0.3303

Summary: ■ ISO 26262-11:2018(E) $\lambda_{\pi\tau}$ 1, element element , element $\times \times \times () \times \lceil \lfloor \lfloor \lfloor \rfloor \rfloor \rfloor \rfloor - \times = \sum N e a t i i i 0$
 $35\ 1\ 3\ element\ FIT = 6,05 = - - - \sum = \times + \times + \times () \times 1\ 3\ 6\ 4\ 4\ 7\ 53\ 10\ 1\ 02\ 10\ 4\ 96\ 10\ , , , 10\ FIT -4\ (13)$ For the λ_2 related term of the die failure rate we get: $Max\ \lambda\ \lambda\ \lambda\ 2, element\ 2, element\ 2, element\ 3\ FIT () = () = () = 2\ 20\ (14)$
 $Max\ t i i i y\ t i i \pi\ \tau\ \pi\ \tau, element\ element, element\ 3 () \times \lceil \lfloor \lfloor \lfloor \rfloor \rfloor \rfloor \rfloor = () \times = \sum 1\ 1\ 01\ 10\ 1\ 1\ 3, \times - = \sum i\ (15)$ $Max\ t i i i y\ \lambda\ \pi\ \tau\ 2, element, element\ element () \times () \times \lceil \lfloor \lfloor \lfloor \rfloor \rfloor \rfloor \rfloor = \sum 1\ 20 \times \times = - 1\ 01\ 10\ 2\ 01\ 1, , FIT\ FIT\ (16)$
This results in an overall die failure rate of: $\lambda_{die}\ FIT\ FIT\ FIT = \times + = - 6\ 05\ 10\ 2\ 01\ 2\ 01\ 4, , , (17)$ To simplify calculation, if the user can identify a match between its product and one of the int

Part: Part 11 - Semiconductors

Clause: Unknown

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Score: 0.3283

Summary: ■ ISO 26262-11:2018(E) Table 30 — Example of failure modes for digital components Part/subpart Function Aspects to be considered for Failure modea Central Processing Unit (CPU) Execute given instruction flow according to given In- struction Set Architecture. CPU_FM1: given instruction flow(s) not executed (total omission) CPU_FM2: un-intended instruction(s) flow executed (commission) CPU_FM3: incorrect instruction flow timing (too early/late) CPU_FM4: incorrect instruction flow result CPU_FM1 can be further refined if necessary into: ■ CPU_FM1.1: given instruction flow(s) not executed (total omission) due to program counter hang up ■ CPU_FM1.2: given instruction flow(s) not executed (total omission) due to instruction fetch hang up CPU Interrupt Handler circuit (CPU_INTH) Execute interrupt

Part: Unknown

Clause: Unknown

Page: 145

Score: 0.3268

Summary: — Data SRAM: Memory where software elements store their own private variables. It also contains communication buffers between software and DMA and between software elements themselves. — Code ROM: Read-only Memory containing the code that is executed by the software elements and possibly constant data used by the software elements. — Software Elements: In this example three software elements are listed: software1, software2 and software3. — Watchdog Interface: It enables to communicate with an external watchdog hardware element. — Shared Resources: The following shared resources are identified: — DMA (Direct Memory Access) hardware element: The DMA can be used by each software element and has read and write access to any addressable resource (Memory, Configuration Register). — EVR (Embedde

Part: Part 11 - Semiconductors

Clause: 1.1

Page: 140

Score: 0.3254

Summary: ■ ISO 26262-11:2018(E) Annex A (informative) Example on how to use digital failure modes for diagnostic coverage evaluation A.1 Example of evaluation of a DMA safety mechanism A.1.1 Description of the use case The following is the DMA use case considered in this example: ■ a message is received by a communication peripheral every X ms; ■ as soon as the message is received by the communication peripheral, it triggers a DMA request; ■ the DMA transfers the message from the peripheral receive buffer to a RAM region; ■ the transfer is always to the same RAM region, independent from the message content; ■ after the DMA is finished with the

transfer, it triggers a CPU interrupt; and ■ the CPU copies the message into a different buffer within the RAM depending on the message ID. A.1.2 Description

Hazard / ASIL

CPU instruction flow omission causing vehicle control failure

ASIL: B

Safety Goal / FSR / TSR

Safety Goal:

Prevent vehicle control failure due to CPU instruction flow omission

FSR:

DMA memory protection unit restricts write access to destination addresses

TSR:

Verify that DMA transfers are protected against unauthorized memory writes

BDD Scenario / Given-When-Then

TC-001 - Verify DMA memory protection during message transfer

Given: A message is received by the communication peripheral every X ms

When: DMA transfers the message from the peripheral receive buffer to a RAM region

Then: The memory protection unit restricts write access to the destination addresses

Expected Action: System triggers safe state if unauthorized access detected

Verification Method: Check memory protection logs for access violations

Methodology: BDD

Traceability

Development Method: BDD

Requirement: REQ-001

Hazard: HZ-001

Safety Goal: SG-001

FSR: FSR-001

TSR: TSR-001

Test Cases: TC-001

Trace Chain:

REQ-001 → HZ-001 → SG-001 → FSR-001 → TSR-001 → TC-001